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Review Article

A systematic review of the roles of agentic AI in higher education

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ABSTRACT

Artificial intelligence in higher education has progressed from reactive tools and generative AI to goal-directed agentic systems capable of autonomous goal pursuit, multi-step planning, tool use, and persistent memory. This review maps the specific roles that agentic AI assumes in higher education and clarifies how these differ from generative and rule-based AI; although prior reviews have examined generative and general AI applications, none has focused specifically on the distinct roles of agentic AI in this setting. Following PRISMA 2020 guidelines, five databases (Scopus, Web of Science, ERIC, Google Scholar, and Wiley Journals) were searched for records published between January 2023 and March 2026, combining agentic, autonomous, and multi-agent AI terminology with higher education descriptors. From 1247 identified records, 843 remained after deduplication; screening yielded 48 included studies subject to thematic synthesis and quality appraisal. Thematic analysis identified six discrete roles: personalised learning and adaptive tutoring; automated assessment and feedback; teaching assistance and augmentation; administrative and student support; curriculum design and workforce alignment; and research support and academic operations. Reported outcomes include academic gains of 15–25%, engagement increases of up to 40%, and substantial reductions in grading time, alongside critical risks of cognitive dependency, cultural homogenisation, algorithmic bias, and inadequate faculty preparedness. Realising the benefits of agentic AI while containing its risks requires a tripartite institutional response: developing AI literacy, establishing ethical infrastructure, and delivering competence-based educator training. Longitudinal empirical studies are needed to establish causal claims about learning outcomes.

1. Introduction

1.1. The evolution of AI in educational contexts

Artificial intelligence entered higher education incrementally, initially through narrow-purpose tools, automated plagiarism detectors, grammar checkers, and rules-based chatbots that responded to predefined queries within strictly bounded domains. These early systems were reactive: they awaited input, matched it against coded patterns, and returned templated outputs. Their value was real but bounded; they automated discrete, low-complexity tasks without any capacity to pursue broader educational goals. The subsequent arrival of machine learning-driven recommendation systems and, later, large language models introduced a qualitatively different capability: the ability to generate contextually sensitive language at scale. Generative AI tools, widely adopted by students and institutions from 2023 onwards, extended the reach of automated support considerably, enabling text generation, summarisation, and conversational interaction. Yet generative AI remained fundamentally reactive, completing tasks presented to it rather than initiating, sequencing, and managing task chains autonomously. Although the roots of educational agentic systems can be traced to the Intelligent Tutoring Systems (ITS) of the 1990s and early 2000s, which could adapt instruction to a learner's performance, those systems lacked the language capabilities, multi-step planning, and external tool use that characterise today's agentic AI. The public release of ChatGPT in November 2022 was a decisive turning point: it made large language models broadly accessible and, indirectly, supplied the reasoning infrastructure on which agentic systems were subsequently built. This is why the present review's search window opens in January 2023 (Section 2.2), the point at which an agentic-AI literature specific to higher education begins to emerge.

The current frontier is marked by the emergence of agentic AI: systems that go beyond responding to prompts and actively pursue goals, decompose those goals into sequences of sub-tasks, call external tools, maintain persistent memory across sessions, and self-adjust in response to changing circumstances (Hosseini & Seilani, 2025). This progression

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from Copilot, in which AI assists a human operator who retains moment-to-moment control, to Autopilot, in which AI manages complex workflows with minimal human intervention, represents a structural shift in the relationship between humans and AI within educational settings (Hosseini & Seilani, 2025). The implications for universities, spanning teaching, assessment, administration, curriculum design, and research, are substantial and only beginning to be understood.

1.2. Defining agentic AI

Agentic AI denotes a class of system architectures rather than a single technology characterised by several intersecting properties. Autonomy is the foundational attribute: the agent executes tasks and makes decisions without requiring human instruction at each step (Hosseini & Seilani, 2025). Reactivity describes the agent's capacity to perceive its environment, including data streams, user inputs, and sensor signals, and to adjust its behaviour accordingly. Proactivity distinguishes agentic systems from purely reactive tools: the agent initiates actions based on inferred goals rather than waiting for explicit commands. Learning ability, typically implemented through reinforcement learning or continual fine-tuning, allows the agent to improve its performance over time by drawing on accumulated experience. Tool use enables the agent to invoke external resources, including search engines, code executors, scheduling systems, and databases, extending its effective capability far beyond the language model at its core. Memory, both short-term within a session and long-term across sessions, allows the agent to maintain contextual continuity, track learner progress, and personalise interactions over weeks or months. Planning, the capacity to decompose complex goals into ordered sub-goals and to sequence actions strategically, is what enables agentic systems to manage multi-step, multi-horizon tasks (Acharya et al., 2025).

These properties collectively enable agents to function as what Acharya et al. (2025) describe as a "qualitative leap" (p. 18912) beyond conventional AI: systems that plan, use tools, maintain memory, and self-adjust to accomplish multi-step, multi-goal tasks that no prior AI paradigm could manage. In contrast to single-task rule engines, agentic AI systems set sub-goals, sequence actions, and can coordinate across multiple specialised agents to pursue long-term objectives (Acharya et al., 2025).

It is important to distinguish agentic AI from generative AI and from traditional autonomous systems. Generative AI creates content in response to prompts but lacks goal-directed planning, tool invocation, and persistent memory as first-class features. Traditional autonomous systems, such as automated grading engines or rules-based scheduling tools, execute fixed algorithms without the adaptive reasoning that characterises agency. Agentic AI subsumes and extends both, combining the language and reasoning capabilities of large language models with goal-directed planning, memory persistence, and real-world tool integration (Hosseini & Seilani, 2025). Table 1 illustrates how these distinctions manifest in higher education deployments.

Table 1
Comparison of rule-based, generative, and agentic AI in higher education.

Dimension	Rule-based/ traditional autonomous AI	Generative AI	Agentic AI
Core capability	Executes fixed, pre-programmed rules	Generates content in response to prompts	Pursues goals autonomously through planning and action
Autonomy	None (deterministic)	Low (responds to each prompt)	High (acts without step-by-step human instruction)
Planning	None	None (single-turn)	Decomposes goals into ordered sub-tasks
Tool use	Fixed integrations only	Generally none	Invokes external tools (search, databases, code executors, schedulers)
Memory	Stateless	Limited to the current session	Persistent across sessions
Higher-education example	Rules-based FAQ chatbot; automated plagiarism check	Essay-drafting or summarisation assistant	Adaptive tutor that plans, monitors progress, and intervenes across a course
Governance implication	Low; bounded and predictable	Moderate; output quality and integrity	High; autonomy raises accountability, privacy, and oversight demands

1.3. Agentic AI in higher education: the research gap

A growing body of secondary research has examined AI in higher education, including systematic literature reviews (SLRs), broader-scope reviews, and bibliometric analyses. Castillo-Martínez et al. (2024) reviewed AI applications broadly across the education sector; Belkina et al. (2025) synthesised case studies of generative AI implementation; Luo et al. (2025) assessed the design and evaluation of AI-based learning tools; and Lan and Zhou (2025) conducted a qualitative review of AI-empowered self-regulated learning. At the bibliometric level, Saleem and Irfan (2026) mapped publication and thematic trends in agentic AI in business higher education using the student life-cycle model and the teaching triad. These contributions are valuable and well cited, and they establish that the wider field is maturing rapidly. More recent syntheses extend this base into adjacent and emerging areas: Kovari (2025) reviewed AI-powered collaborative learning, and Bittle and El-Gayar (2025) synthesised the evidence on generative AI and academic integrity. Forward-looking and agent-specific work has also begun to appear. Sudarshan et al. (2026) outline a multi-agent ecosystem perspective for higher education, while Andalusia et al. (2026) propose a personality-aware agentic AI framework for academic and career advising. However, while previous reviews have examined generative AI and general AI applications, none has specifically focused on the distinct roles of agentic AI, namely autonomous, goal-directed systems with planning, tool use, and persistent memory, across the full range of university functions, or on how those roles differ in terms of function, pedagogy, risk, and governance. Existing bibliometric work, such as Saleem and Irfan (2026), maps publication patterns but is confined to business education and does not synthesise the substantive roles agentic AI performs. Our contribution is therefore to clarify, rather than deny, where prior work stops and ours begins: this gap matters because the governance, ethical, and pedagogical implications of agency are categorically different from those of reactive AI tools. Complementing these, Liu et al. (2026) systematically review generative-AI agents in education, mapping macro-level trends, distinct educational agent roles, and their theoretical foundations, while Mena Guacas et al. (2026), writing in this journal, bibliometrically map artificial-intelligence-driven innovation across higher education. Understanding the specific roles agentic AI plays—rather than only its technical features—matters because different roles carry different pedagogical assumptions, risk profiles, and governance requirements: an autonomous tutor, an automated grader, and a research agent each raise distinct questions of oversight and accountability.

Sargsyan (2025) and Joshi (2026) both note substantial empirical gaps in the field and call for more rigorous investigation, particularly regarding long-term learning outcomes and ethical governance. Kasarci et al. (2025) similarly identify the lack of systematic frameworks for managing AI ethics in institutional contexts. The present review addresses this lacuna directly by synthesising evidence on the specific roles agentic AI assumes in higher education, the theoretical frameworks that underpin those roles, the benefits and challenges reported in the literature, and the governance considerations that emerge. This gap is addressed directly through four research questions that examine what roles agentic AI assumes (RQ1), which theoretical frameworks underpin

those roles (RQ2), what benefits, risks, and implementation challenges are reported (RQ3), and what governance and ethical considerations emerge (RQ4). The need is practical as well as scholarly: as agentic systems are increasingly piloted and deployed across universities, policymakers, institutional leaders, and educators require a synthesis capable of informing institutional strategy, faculty development, and ethical governance before these systems become entrenched without adequate oversight.

This review makes three contributions. Theoretically, it organises a fragmented literature into a role-based framework and links each role to the pedagogical and learning theories that justify it (notably Vygotsky's Zone of Proximal Development, self-regulated learning, cognitive load theory, Bloom's taxonomy, and critical pedagogy), thereby distinguishing agency from generative capability at a conceptual level. Contextually, it focuses on higher education specifically (rather than education broadly or a single discipline such as business) and foregrounds equity and Global South considerations that are underrepresented in prior reviews. Methodologically, it applies a transparent, PRISMA 2020-compliant thematic synthesis with disclosed search strings, inclusion criteria, dual screening, and MMAT/CASP quality appraisal, complementing the bibliometric approach of Saleem and Irfan (2026) with a substantive synthesis of what agentic systems actually do. Unlike Liu et al. (2026), who typologise generative-AI agents, the present review restricts its unit of analysis to systems meeting the agentic criteria defined in Section 2.3—autonomy, planning, tool use, and persistent memory—yielding a role taxonomy of autonomous rather than merely reactive agents.

The remainder of the paper is structured as follows. Section 2 details the PRISMA-based methodology, including the search strategy, inclusion and exclusion criteria, screening, and quality appraisal. Section 3 reports the six agentic AI roles identified through thematic synthesis. Section 4 discusses the theoretical foundations, benefits, risks, and governance and ethical considerations, and states the limitations of the review. Section 5 concludes with principal findings, a tripartite framework for responsible integration, and directions for future research. To guide this synthesis, the review pursues the objective of mapping the roles of agentic AI in higher education and is organised around four research questions.

1.4. Research questions

This review is organised around four research questions:

RQ1 What roles does agentic AI assume in higher education contexts?
RQ2 What pedagogical frameworks and theoretical foundations underpin agentic AI integration?
RQ3 What are the reported benefits, risks, and implementation challenges of agentic AI in higher education?
RQ4 What governance and ethical considerations emerge from the literature on agentic AI in higher education?

2. Methodology

2.1. Review design and protocol

This systematic review follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines (Page et al., 2021). The review was designed to produce a comprehensive thematic synthesis of empirical and theoretical literature concerning the roles of agentic AI in higher education. A protocol was established before data collection, specifying the search strategy, inclusion and exclusion criteria, data extraction procedures, synthesis approach, and quality assessment method. Thematic analysis was selected as the primary synthesis strategy given the diversity of study designs in the included literature, which spans experimental studies, design-based research, systematic reviews, and theoretical frameworks. The PRISMA 2020 flow diagram is presented as Fig. 1. The review protocol was established a priori but was not prospectively registered (for example, in PROSPERO).

2.2. Search strategy

Five databases were searched: Scopus, Web of Science, ERIC, Google Scholar, and Wiley Online Library. The search covered publications from January 2023 to March 2026, a period chosen to capture the rapid emergence of agentic AI systems following the widespread adoption of large language models from late 2022. Earlier publications were excluded because the terminology, architectures, and applications characteristic of contemporary agentic AI were not yet established in the education literature.

The following Boolean search string was applied, with minor adaptations for database-specific syntax:

("agentic AI" OR "AI agents" OR "autonomous AI" OR "multi-agent systems") AND ("higher education" OR "university" OR "tertiary education" OR "post-secondary")

Records retrieved by database were as follows: Scopus (312), Web of Science (287), ERIC (198), Google Scholar (385), and Wiley Journals (65), yielding a combined total of 1247 records.

To ensure currency at revision and to broaden term coverage, the search was updated in July 2026 using an expanded set of synonyms on the AI side ("agentic artificial intelligence," "LLM agents," "autonomous agents," "intelligent agents," and "generative agents") and on the higher-education side ("college," "undergraduate," and "postgraduate"). The primary corpus of 48 studies was held at the prespecified March 2026 cut-off to preserve the integrity of the screening counts reported below; eligible sources identified in the update, including the most recent 2026 systematic and bibliometric reviews, are integrated narratively in the Discussion, Limitations, and Future Research sections rather than re-entered into the quantitative screening flow.

2.3. Inclusion and exclusion criteria

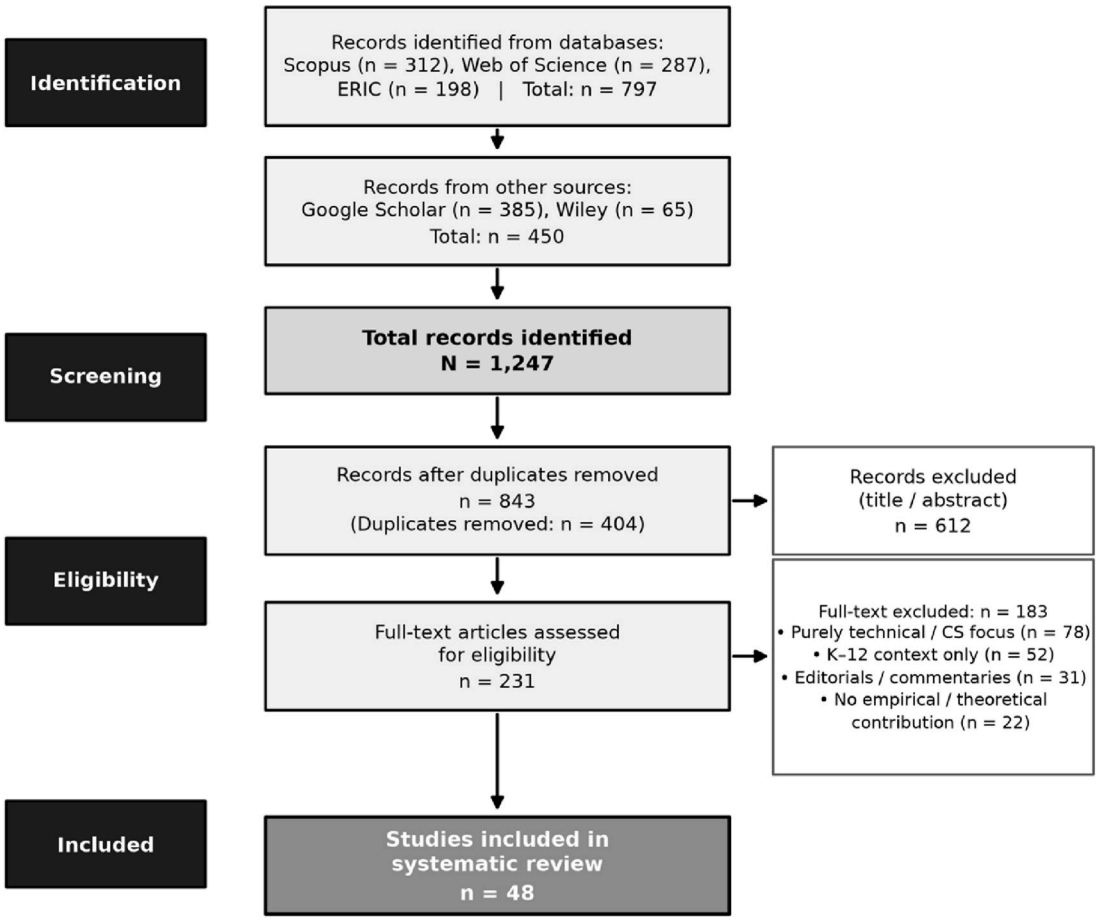
Inclusion and exclusion criteria were specified at protocol stage and are presented in Table 2.

To operationalise the AI-focus criterion and to distinguish agentic AI from generative or rule-based systems, four defining features were applied, consistent with the definition in Section 1.2: (1) autonomy in decision-making without step-by-step human instruction; (2) planning, that is, the decomposition of goals into ordered sub-tasks; (3) use of external tools (for example, search engines, databases, code executors, or scheduling systems); and (4) persistent memory across interactions or sessions. A study was classified as agentic when the system it described met at least three of these four criteria; systems meeting fewer than three were treated as generative or rule-based and excluded on AI-focus grounds.

In addition to the peer-reviewed corpus, a small number of industry and grey-literature sources (for example, vendor platform documentation and technology reports) are cited illustratively in the Results and Discussion to describe real-world deployments. These sources fall outside the 48 peer-reviewed included studies, are not counted among them, and are not treated as primary evidence of effectiveness; figures drawn from them are reported as vendor- or developer-reported and interpreted with corresponding caution.

2.4. Screening process

After deduplication using Zotero reference management software, 843 unique records remained. The author and an independent second screener (a colleague not listed as an author) screened titles and abstracts against the inclusion and exclusion criteria; disagreements were



Note. PRISMA = Preferred Reporting Items for Systematic Reviews and Meta-Analyses (Page et al., 2021).

Fig. 1. PRISMA 2020 flow diagram illustrating the systematic search and screening process. Records were identified from Scopus (312), Web of Science (287), ERIC (198), Google Scholar (385), and Wiley Journals (65), totalling 1247 records. After deduplication (n = 843), title and abstract screening excluded 612 records, leaving 231 for full-text review. A further 183 were excluded after full-text assessment, yielding 48 included studies.

Table 2
Inclusion and exclusion criteria.

Criterion	Inclusion	Exclusion
Publication period	January 2023 to March 2026	Prior to January 2023
Publication type	Peer-reviewed journal articles; peer-reviewed conference papers	Editorials, commentaries, opinion pieces without empirical or theoretical contribution
Education level	Higher education (university, college, post-secondary, tertiary)	Studies focused exclusively on K-12; studies with no clear HE component
AI focus	Studies foregrounding agentic, autonomous, or multi-agent AI systems	Studies focused on generative AI without agentic properties; studies on rule-based AI without autonomy or planning
Contribution type	Empirical findings (quantitative, qualitative, or mixed methods) or substantive theoretical contribution	Purely technical or computer science papers without education application
Language	English	Non-English publications

resolved through discussion until full consensus was reached. This stage excluded 612 records: studies not relevant to the HE context (n = 289), studies not peer-reviewed (n = 187), and studies not focused on agentic or autonomous AI (n = 136). The remaining 231 records proceeded to full-text review.

Full-text assessment excluded a further 183 articles: purely technical computer science papers without education focus (n = 78), studies concerning K-12 only (n = 52), editorials and commentaries (n = 31), and studies with no empirical or theoretical contribution (n = 22). The final corpus comprised 48 included studies. The complete PRISMA 2020 screening flow is illustrated in Fig. 1.

2.5. Data extraction and thematic synthesis

Data extraction was conducted using a structured coding template capturing: study citation, study type and design, country and institutional context, AI system described (including agentic properties), education context (discipline, level, setting), reported outcomes and findings, theoretical frameworks cited, and limitations identified by the authors. Data were extracted by the author, and the independent second

screener cross-checked a 15% random sample for reliability.

Codes were grouped into thematic categories corresponding to the roles that agentic AI was found to play within higher-education contexts; these roles are reported in Section 3. Coding and data management were carried out in NVivo (version 14), whose framework-matrix functionality supported the charting stage of the framework approach and the systematic organisation of codes across studies.

2.6. Quality assessment

Study quality was appraised using the Mixed Methods Appraisal Tool (MMAT) version 2018 (Hong et al., 2018), which provides criteria for qualitative, quantitative, and mixed-methods studies. Theoretical and review-type studies were assessed using criteria adapted from the Critical Appraisal Skills Programme (CASP) systematic review checklist. Quality scores were recorded descriptively and used to weight the confidence placed in specific findings during synthesis; no studies were excluded solely on quality grounds, but low-quality studies are noted where cited in evidence chains. A summary of included study characteristics is presented in Table 4.

Table 3 summarises the key methodological characteristics of the review, allowing the full protocol to be seen at a glance.

3. Results

3.1. Descriptive overview of included studies

The 48 included studies were published between 2023 and early 2026, with a pronounced acceleration in output from 2024 onwards (see Fig. 2). Studies from 2023 constituted 10% of the corpus (five studies), those from 2024 accounted for 29% (14 studies), those from 2025 constituted 44% (21 studies), and those from early 2026 made up the remaining 17% (eight studies). This trajectory reflects the rapid expansion of agentic AI as an operational and research focus in higher education following the widespread availability of large language model platforms.

Geographically, the corpus was dominated by studies from North America (35%) and East and South Asia (28%), with European contributions accounting for 21%, studies from the Global South (including Sub-Saharan Africa and Latin America) comprising 10%, and the remaining 6% classified as multi-regional or international in scope. The under-representation of Global South contexts is a notable limitation of the current evidence base, particularly given the differentiated infrastructure and equity challenges those contexts face (Khoza & van der Walt, 2025). The table of study characteristics (Table 4) summarises study type, geographic origin, methodological approach, and agentic AI application domain for a representative sample of the included studies.

Methodologically, the corpus was diverse. Empirical studies constituted 63% of the total (30 studies), encompassing experimental and quasi-experimental designs (n = 12), qualitative case studies (n = 11), and mixed-methods studies (n = 7). Theoretical and conceptual papers accounted for 17% (eight studies). Review articles, including scoping reviews and secondary analyses, represented 20% (10 studies). The predominance of case studies and design-based research, and the relative scarcity of randomised controlled trials, reflects the early stage of the field. As discussed in Section 4.5, this methodological distribution limits the strength of causal claims that can currently be advanced.

Across the corpus, six discrete thematic roles for agentic AI in higher education were identified: personalised learning and adaptive tutoring;

Table 3
Summary of key methodological characteristics.

Step	Details
Databases	Scopus, Web of Science, ERIC, Google Scholar, Wiley Online Library
Time period	January 2023 – March 2026 (search updated July 2026)
Search string	("agentic AI" OR "AI agents" OR "autonomous AI" OR "multi-agent systems") AND ("higher education" OR "university" OR "tertiary education" OR "post-secondary"); expanded with additional synonyms in the July 2026 update
Inclusion criteria	Peer-reviewed; English language; higher-education context; agentic AI focus (≥3 of 4 agentic criteria)
Total records identified	1247
After deduplication	843
Full-text assessed	231
Final included studies	48
Screening	Two independent reviewers (title/abstract and full text); third-reviewer adjudication of disagreements
Quality assessment	Mixed Methods Appraisal Tool (MMAT 2018); CASP for review/theoretical studies
Synthesis method	Thematic synthesis using the framework approach (Ritchie & Spencer, 1994), managed in NVivo 14

Table 4
Characteristics of a representative sample of included studies.

Study	Year	Country/ Region	Design	AI Role(s) Addressed
Yuensook et al.	2025	Thailand	Systematic review	Personalised learning, adaptive tutoring
Gupta & Heggond	2026	India	Systematic review (PRISMA)	Personalised learning, multi-agent tutoring
Meti	2026	India	Literature review	Personalised learning, autonomous tutoring
López-Goyez et al.	2026	Ecuador	Case study (Moodle)	Adaptive tutoring, RL architectures
Rumble et al.	2026	UK	Design and simulation	Personalised learning (LEA framework)
Sargsyan	2025	Armenia	Conceptual	Teaching assistance, ethics, governance
Yan et al.	2025	Australia/ China	Experimental	Teaching assistance, visual analytics
Simmhan & Kulkarni	2025	India	Case study (IISc)	Teaching assistance, instructor agents
Joshi	2026	USA	Conceptual/ policy	Curriculum design, workforce alignment
Kasarci et al.	2025	Turkey	Systematic review	Governance and ethics
Lan & Zhou	2025	China/intl.	Qualitative systematic review	Self-regulated learning, pedagogy
Luo et al.	2025	Singapore	Systematic review	AI-based learning tools, assessment
Belkina et al.	2025	Australia	Systematic review	GenAI implementation, pedagogy
Castillo-Martínez et al.	2024	Mexico/ Spain	Systematic review	AI in higher education broadly
Khoza & van der Walt	2025	South Africa	Systematic review	Global South, pedagogy
Msafiri et al.	2025	Tanzania/ intl.	Empirical	Collaborative learning environments
Roe & Perkins	2024	UK/ Australia	Scoping review	Agency in education, GenAI
Güneş & Liman Kaban	2025	Turkey	Delphi study	Academic integrity, ethics
Gao & Tan	2025	China	Systematic review	Undergraduate/ vocational HE
Acharya et al.	2025	USA	Conceptual review	Agentic AI roles broadly
Hosseini & Seilani	2025	Iran	Systematic review	Agentic AI characteristics, Copilot/Autopilot

Note. This table presents a representative sample of the 48 included studies (those cited in Sections 3–4); the complete study-level list is available from the corresponding author on reasonable request.

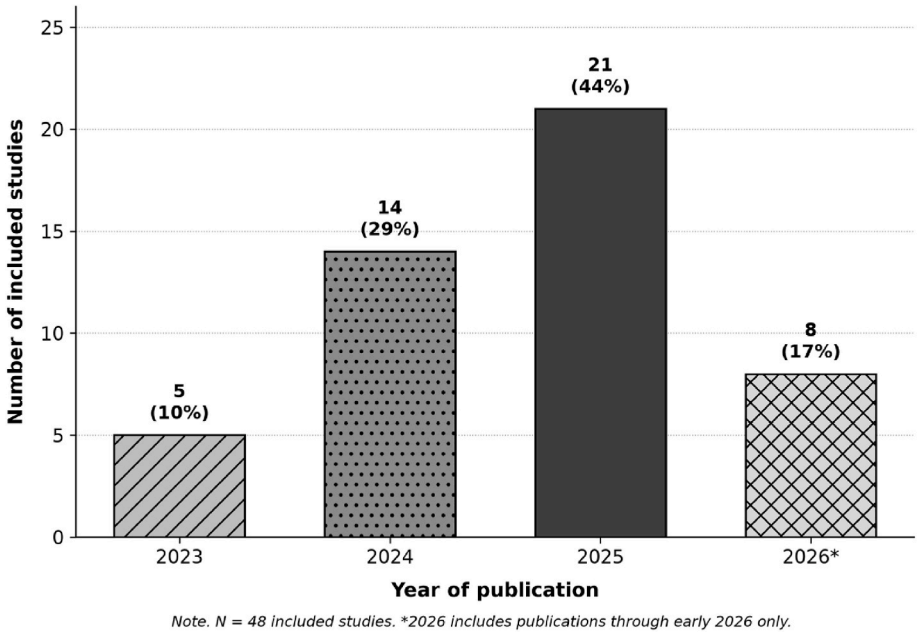


Fig. 2. Publication trends of agentic AI in higher education research, 2023–2026. The bar chart illustrates the accelerating growth in peer-reviewed output across the search period: five studies (2023), 14 studies (2024), 21 studies (2025), and eight studies (early 2026).

automated assessment and feedback; teaching assistance and augmentation; administrative and student support; curriculum design and workforce alignment; and research support and academic operations. These roles are not mutually exclusive in practice, and several studies reported systems serving more than one function. Table 5 presents the thematic categories, characteristic system types, representative studies, and frequency of appearance in the corpus. Fig. 3 maps the thematic relationships between roles and the cross-cutting concerns of governance, ethics, and pedagogy.

Table 5
Thematic categories and representative studies.

Theme	Role Description	Representative Studies	n Studies
Personalised learning and adaptive tutoring	Autonomous, adaptive learning path generation; intelligent tutoring agents	Gupta and Heggond (2026); Meti (2026); Yuensook et al. (2025); López-Goyez et al. (2026); Rumble et al. (2026)	18
Automated assessment and feedback	Rubric-based grading agents; real-time feedback generation; automated marking	Luo et al. (2025); Gao and Tan (2025)	12
Teaching assistance and augmentation	Virtual TA functions; AI instructor agents; content delivery support	Simmhan and Kulkarni (2025); Yan et al. (2025); Sargsyan (2025); Acharya et al. (2025)	11
Administrative and student support	Enrolment agents; early warning systems; advising and scheduling	Joshi (2026); Msafiri et al. (2025)	8
Curriculum design and workforce alignment	AI-driven curriculum redesign; workforce competency mapping	Joshi (2026); Gao and Tan (2025)	5
Research support and academic operations	Literature review automation; multi-agent research frameworks	Acharya et al. (2025); Hosseini and Seilani (2025)	4

3.2. Role 1: personalised learning and adaptive tutoring

Personalised learning and adaptive tutoring constitutes the most extensively represented role in the reviewed corpus, appearing in 18 of 48 studies. At its core, this role involves agentic AI systems that continuously model individual learner states, dynamically adjust instructional content and pacing, and provide targeted feedback without requiring constant human orchestration. The shift from static, one-size-fits-all instruction to genuinely individualised learning at scale is the central promise driving research and deployment in this area.

The technical foundations of adaptive tutoring agents rest on reinforcement learning, natural language processing, affective computing, and cognitive modelling. Gupta and Heggond (2026) provide a comprehensive review of multi-agent cognitive architectures for personalised adaptive learning, documenting how specialised agents for profiling, content selection, difficulty adjustment, and engagement monitoring operate in coordinated configurations. Their analysis demonstrates that multi-agent systems exhibit superior behaviour modelling, more fine-grained organisation of individualised learning processes, richer real-time feedback, and stronger metacognitive enhancement capacities compared to traditional single-agent or rule-based tutoring technologies. Meti (2026) independently converges on similar conclusions, noting advances in real-time knowledge tracking, emotion-sensitive pedagogy, and optimal curriculum sequencing as defining characteristics of agentic tutoring architectures.

The López-Goyez et al. (2026) Moodle-based case study provides a concrete illustration of how reinforcement learning and generative AI can be integrated within an existing learning management system. The ELA Tutor architecture they describe demonstrates self-adaptive behaviour, adjusting instructional difficulty and scaffolding in real time as students interact with course materials. This integration with institutional infrastructure is significant: it addresses one of the persistent barriers to agentic AI adoption by embedding autonomous capability within existing platforms rather than requiring wholesale system replacement.

Rumble et al. (2026) present the Learning Engagement Assistant (LEA), a tri-modal agentic AI framework developed at Abertay University. LEA operates through three integrated modes (Chat, Tutor, and Quiz), orchestrated by an adaptive agent that continuously monitors

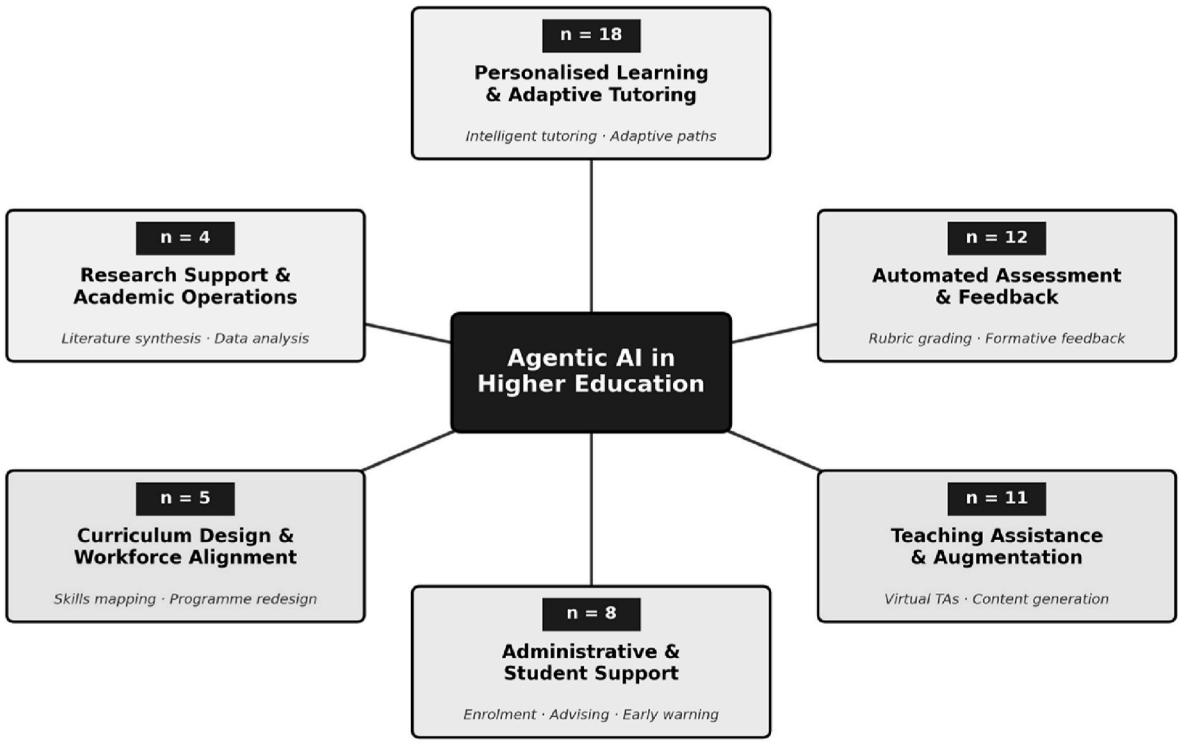


Fig. 3. Thematic map of agentic AI roles in higher education. A hub-and-spoke diagram illustrating the six identified roles and their relationships to the central organising theme. The value n beside each role denotes the number of included studies in which that role appeared; roles are not mutually exclusive.

cognitive load, infers each learner's Zone of Proximal Development, and adjusts scaffolding and content difficulty accordingly. LEA's architecture combines course-specific Retrieval-Augmented Generation (RAG) with a Knowledge Component model to provide pedagogically grounded instruction aligned with learning objectives. Simulation-based evaluation of LEA demonstrated robust retrieval accuracy, coherent multi-turn tutoring, and adaptive stability across diverse learner profiles, confirming that a multi-agent, tri-modal architecture can sustain personalised instruction at scale. The LEA framework operationalises Vygotsky's Zone of Proximal Development, cognitive load theory, and self-efficacy modelling within a unified agentic architecture, as discussed further in Section 4.1.

Platforms such as Khanmigo and Duolingo's AI-enhanced scaffolding have demonstrated real-world deployment of scenario-based adaptive tutoring at scale, providing students with real-time feedback calibrated to their performance without instructor intervention (Acharya et al., 2025; Sargsyan, 2025). These platforms exemplify the shift from reactive question-answering to proactive instructional management, where the agent initiates review sequences, diagnoses misconceptions, and personalises the pacing of progression.

The quantitative evidence on outcomes is encouraging, though methodological limitations temper strong causal claims (see Section 4.5). Yuensook et al. (2025), in their systematic review of AI-driven adaptive learning systems in higher education, report academic performance gains of 15–25% and improved learner engagement of up to 40% across the studies included in their synthesis. These figures represent the most comprehensive outcome aggregation currently available and align directionally with findings from individual studies. However, Yuensook et al. (2025) also document challenges including insufficient technical infrastructure, faculty readiness gaps, ethical concerns, and the under-representation of non-STEM disciplines in the evidence base. The overwhelming focus of adaptive tutoring research on STEM and language learning contexts leaves significant questions about generalisability to the humanities, social sciences, and professional education domains.

3.3. Role 2: automated assessment and feedback

The role of agentic AI in automated assessment and feedback is the second most prominent theme in the reviewed corpus, appearing in 12 studies. Assessment automation encompasses rubric-based grading, automated essay scoring, formative feedback generation, and real-time performance diagnostics. The appeal is evident: assessment constitutes a substantial burden for instructors, with manual grading consuming significant professional time, while delayed feedback is well-established as a hindrance to learning (Luo et al., 2025).

Contemporary agentic assessment systems go considerably beyond earlier keyword-matching tools. Modern systems integrate NLP-based semantic analysis, optical character recognition for handwritten submissions, machine learning models trained on large corpora of graded work, and rubric-following agents that apply institutional criteria consistently across submissions. These capabilities have produced reported grading time reductions of 50–70%, with AI graders demonstrating 90–95% agreement with human graders across multiple studies.

The adoption of automated assessment is not without contested terrain. Two significant critical concerns appear consistently in the literature. First, automated grading systems can introduce bias against non-native English speakers when training data are insufficiently

diverse and representative. Algorithmic scoring systems may penalise linguistic features associated with non-standard English varieties, cultural rhetorical conventions, or multilingual code-switching, producing systematically inequitable outcomes for international and non-Western students. Second, Sargsyan (2025) raises a concern that cuts to the pedagogical heart of assessment: that passive reliance on automated grading risks eroding students' metacognitive development. Assessment, in Sargsyan's analysis, functions as a site of reflective learning and not only as a gatekeeping mechanism. When automated systems return scores without fostering genuine self-evaluation, critical review, or engagement with the assessment process itself, students may be deprived of a key mechanism through which deep learning occurs. Sargsyan (2025) advocates explicitly for learning designs that embed metacognitive prompts within automated feedback workflows, ensuring that agentic feedback functions as an invitation to further reflection rather than a closed verdict. Luo et al. (2025) similarly call for designs that maintain student agency within assessment contexts, noting that the transition to AI-mediated evaluation requires deliberate pedagogical scaffolding to preserve its educational integrity.

Gao and Tan (2025) contribute a differentiated perspective, noting that the uptake and form of automated assessment varies meaningfully between undergraduate and vocational higher education contexts. Vocational programmes, with their emphasis on competency demonstration and practical skill evaluation, present distinct assessment challenges that may require different agentic configurations from those developed primarily for academic essay scoring. This differentiation underscores the importance of context-sensitive deployment design rather than assuming that assessment agents effective in one domain will transfer straightforwardly to another.

3.4. Role 3: teaching assistance and augmentation

The role of agentic AI as a virtual teaching assistant and instructor augmentation tool appears in 11 studies, making it the third most prominent theme. This role spans a spectrum from narrow task automation, such as answering routine administrative questions, to full instructional delivery as the primary course instructor, with important implications for conceptions of educator identity and practice at each point along that spectrum.

At its most common deployment, agentic AI serves as a virtual teaching assistant (TA): answering students' routine questions about course logistics, retrieving learning materials on request, tracking individual learner progress, flagging unresolved queries for escalation to human instructors, and providing immediate feedback on low-stakes formative exercises (Acharya et al., 2025; Hosseini & Seilani, 2025). This configuration positions agentic AI as relieving educators of administrative and procedural burden while keeping human instructors central to high-order pedagogical decision-making, mentorship, and complex student engagement. The potential for 24/7 availability is significant in large-scale and online programmes, where synchronous access to instructors is structurally constrained.

The most substantive empirical investigation of full agentic instruction appears in Simmhan and Kulkarni's (2025) study at the Indian Institute of Science (IISc), Bangalore. Their AI Instructor Agent framework, implemented within Microsoft Teams Copilot and integrated with Moodle, positioned an LLM-driven agent as the primary instructor in a graduate-level Cloud Computing course, with the human instructor adopting a facilitating role, providing course structure, curriculum design, and end-of-session question-and-answer interaction. The agent was configured with course-level scaffolding, weekly topic-specific knowledge bases aligned to Bloom's taxonomy, and pedagogical guardrails grounded in the Knowledge-Learning-Instruction (KLI) framework. Three design principles governed agent behaviour: pedagogical specificity (constraining each agent to its assigned module), context persistence (retaining dialogue history across sessions), and engagement mediation (balancing reactive response with proactive initiative such as brief quizzes). Engagement analytics were generated automatically from student-submitted chat transcripts, providing metrics of topic coverage, topic depth, and turn-level elaboration. The study reported a shift in engagement patterns across successive modules, from broad conceptual exploration in early weeks to deeper, more focused inquiry in later weeks, suggesting that agentic instruction can sustain and deepen student engagement over time. This research constitutes one of the first field deployments of an agentic AI as primary instructor rather than supplementary tool.

Yan et al. (2025) contribute an experimental study investigating how generative AI agents and scaffolding enhance students' comprehension of visual learning analytics. Their findings illustrate the potential of AI agents to function as interpretive mediators between complex data visualisations and student understanding, a role that is increasingly significant as learning analytics dashboards become standard features of institutional infrastructure.

The literature consistently argues that the augmentation of teaching by agentic AI reconfigures the responsibilities of human educators rather than displacing them. Hosseini and Seilani (2025) propose that as agentic AI absorbs procedural and administrative teaching tasks, human instructors can concentrate on facilitating conceptual discussion, nurturing critical thinking, providing ethical judgment, and offering the kind of emotionally attuned mentorship that AI systems cannot yet replicate. Sargsyan (2025) urges caution, however, noting that without institutional frameworks for AI literacy and governance, instructors risk being sidelined from critical pedagogical decisions, with agentic systems filling vacuums that should properly remain under human professional authority. Realising this reconfiguration in practice depends heavily on instructor uptake. Chen et al. (2025), in a human-centred design study of an interactive pedagogical agent, find that adoption hinges on trust, social transparency, and instructors' ability to control how much they engage with the system. These factors are especially salient for AI-cautious faculty and, if neglected, may widen rather than narrow pedagogical divides.

3.5. Role 4: administrative and student support

Administrative and student support functions constitute the fourth identified role, appearing in eight studies. This role encompasses two distinct but related application domains: the automation of institutional administrative processes (enrolment, scheduling, advising, and document management) and the deployment of proactive student support agents designed to identify at-risk students and provide timely, personalised intervention.

In enrolment and administrative operations, multi-agent platforms have demonstrated substantial operational gains. Element451's Bolt Agent ecosystem, designed specifically for higher education, deploys coordinated agent teams across the student lifecycle, from prospective student engagement and application processing through to enrolment conversion and academic support. Institutions using integrated AI support tools report significant improvements in engagement and follow-through rates (Element451, 2025). The shift from reactive administrative systems, where staff respond to student enquiries and exceptions after they arise, to proactive agentic systems, which continuously monitor lifecycle indicators and intervene before problems escalate, represents a structural change in how institutions manage student relationships.

Early warning systems represent one of the most consequential student support applications in the corpus. These systems apply machine learning models to continuous streams of student data, including LMS login frequency, assignment submission rates, attendance records, advising visit patterns, and financial aid status, to generate real-time risk assessments and trigger targeted interventions. Evidence from across the literature suggests that appropriately designed early warning systems can identify at-risk students. Element451's own platform materials report (vendor-reported) expected retention improvements of 2–7%

term-to-term and grade gains of 3–10% where agents trigger timely interventions (Element451, 2025). Real-world implementations reinforce these findings: Ivy Tech Community College's predictive analytics programme identified roughly 16,000 at-risk students from a population of 60,000 within two weeks; the pilot helped about 3000 of the contacted students achieve a grade of C or better, and 98% of those contacted responded neutrally or positively to the outreach (Google for Education, 2025). The Universitat Oberta de Catalunya reported a 12% reduction in course dropout rates following the deployment of a personalised daily messaging agent for new students. In the Ivy Tech case study, the underlying machine-learning model reportedly predicted course outcomes with approximately 80% accuracy (Google for Education, 2025).

The proactive-reactive distinction carries important pedagogical and ethical dimensions. Proactive intervention systems do not wait for students to request help, they identify risk indicators and act on them before difficulties have become entrenched. This capacity substantially extends the institutional welfare function in contexts where student-to-advisor ratios constrain timely human support. At the same time, the continuous collection of behavioural data on which these systems depend raises data privacy questions, discussed in Section 4.4, concerning consent, transparency, and the potential for surveillance dynamics to undermine trust relationships between students and institutions (Kasarci et al., 2025).

Msafiri et al. (2025) examine the impact of AI on personal and collaborative learning environments and document the capacity of agentic systems to coordinate across both individual support and group learning contexts. Their work notes that AI support which is well integrated into collaborative workflows can preserve and enhance social learning, while poorly designed systems risk fragmenting peer interaction by over-individualising instructional pathways.

3.6. Role 5: curriculum design and workforce alignment

Curriculum design and workforce alignment, though representing only five studies in the corpus, constitutes one of the most strategically significant roles identified. This role concerns the use of agentic AI to analyse labour market data, identify competency gaps, recommend curriculum modifications, and align programme offerings with evolving workforce demands.

Joshi (2026) presents the most detailed treatment of this role, arguing that agentic AI can help institutions redesign curricula by systematically mapping course content against workforce competency frameworks, scanning real-time labour market data, and identifying where graduate preparation diverges from employer demand. In the specific context of financial planning education, Joshi (2026) demonstrates how agentic AI investment firm models have achieved 50–70% cost reductions while maintaining fiduciary standards, establishing the economic imperative for proactive curriculum revision. The curricular framework proposed encompasses foundational AI literacy modules for all students, specialised agentic AI courses with hands-on laboratory components, AI-augmented redesign of core disciplinary courses, interdisciplinary project-based learning with computer science, and governance modules addressing regulatory compliance. This multi-layered integration strategy illustrates how agentic AI can function simultaneously as the subject of curriculum reform (what students learn) and as an instrument for conducting that reform (how institutions redesign programmes).

Gao and Tan (2025) note that the form of AI-driven curriculum alignment differs significantly between undergraduate university education and vocational higher education, with vocational contexts placing greater emphasis on demonstrable competency outcomes and industry alignment. Cross-functional campus agents capable of coordinating between academic planning, student services, and employer relations represent an emergent implementation mode identified in several theoretical papers, though empirical evidence for such integrated deployments remains limited in the reviewed corpus.

The workforce alignment role intersects with broader questions about the purpose and orientation of higher education. Joshi (2026) acknowledges that while AI-driven curriculum alignment with market demand strengthens graduate employability, it must be balanced against the institution's commitments to broad intellectual formation, critical citizenship, and knowledge creation for its own sake. This tension, between responsive vocational preparation and liberal educational mission, is not unique to the AI era but is sharpened by the speed and granularity with which agentic AI can operationalise market signals as curriculum directives.

3.7. Role 6: research support and academic operations

Research support and academic operations constitute the smallest but most technically ambitious role identified in the corpus, appearing in four studies. This role involves agentic AI systems that assist with, or autonomously conduct, components of the research process, including literature review, hypothesis formulation, experimental design, code generation, and report writing.

The most prominent empirical demonstration of this role is the Agent Laboratory framework, developed collaboratively by researchers at Johns Hopkins University and AMD. Agent Laboratory employs a coordinated team of specialised AI agents, configured to mirror real-world research roles (PhD researcher, postdoctoral researcher, machine learning engineer), to progress a user-provided research idea through three structured stages: literature review (using the arXiv API to gather and organise relevant papers), plan formulation, and experimental implementation with automated reporting. In its autonomous mode, the framework demonstrated, in an industry-reported evaluation, an 84% reduction in research costs compared to previous autonomous research methods, with GPT-4o completing the full research workflow in approximately 1165 s at a cost of US$2.33 per paper (Unite.AI, 2025). Human oversight in "co-pilot" mode improved output quality ratings from an average of 3.8/10 to 4.38/10, with particularly significant improvements in clarity (+0.23) and presentation (+0.33). This evidence suggests that the optimal mode of agentic research assistance is structured human-AI collaboration rather than fully autonomous operation, consistent with the augmentation thesis advanced across the broader literature.

The Agent Laboratory example also illustrates the scalability implications of agentic research support. For institutions with limited research capacity, particularly in the Global South or in teaching-intensive universities, the ability to automate resource-intensive early-stage research processes (literature synthesis and protocol development) could meaningfully democratise research participation. At the same time, questions about intellectual provenance, research integrity, and the appropriate attribution of AI-assisted outputs are urgent, as discussed in Section 4.4.

Hosseini and Seilani (2025) identify literature review automation and multi-agent orchestration of research workflows as an important frontier, noting that the integration of agentic research tools with emerging technologies such as quantum computing could unlock capabilities that are currently difficult to anticipate. Acharya et al. (2025) extend this by arguing that memory-rich research agents capable of maintaining context across extended research projects, and of operating within institutionally defined ethical guardrails, represent the next generation of research infrastructure for universities willing to invest in their development. The thematic map in Fig. 3 illustrates how the six identified roles interrelate and how cross-cutting concerns of ethics, governance, and pedagogy extend across all of them.

4. Discussion

The findings reported above speak directly to the four research questions and to the gap identified in Section 1.3. Having mapped the six

roles that agentic AI assumes in higher education (RQ1), this section interprets the theoretical foundations that underpin them (RQ2), weighs the benefits and risks reported across the corpus (RQ3), and examines the governance and ethical considerations they raise (RQ4). In doing so, it connects the role-based taxonomy introduced earlier to the pedagogical, ethical, and governance debates that motivate the review, and returns in Section 5 to the tripartite framework proposed as this review's central contribution.

4.1. Theoretical foundations underpinning agentic AI roles

The integration of agentic AI into higher education draws on a diverse set of theoretical traditions, each of which inflects how specific roles are designed, evaluated, and justified. Understanding these theoretical foundations is essential for assessing the pedagogical coherence of existing systems and for identifying where theoretical grounding is absent or contested.

Vygotsky's sociocultural theory, and in particular the concept of the Zone of Proximal Development (ZPD), provides the most frequently invoked theoretical anchor for adaptive tutoring systems. The ZPD, the region between what a learner can accomplish independently and what they can accomplish with appropriate support, furnishes a principled rationale for dynamic scaffolding: the agent's role is to identify each learner's ZPD in real time and to provide the precise level of support that enables progression without inducing cognitive overload. The LEA framework (Rumble et al., 2026) explicitly operationalises ZPD inference as a core component of its orchestration logic, alongside cognitive load estimation derived from cognitive load theory and motivation tracking grounded in self-efficacy models. This multi-theoretical architecture represents a sophisticated attempt to translate developmental psychology into algorithmic specification.

Self-regulated learning (SRL) theory is the second major theoretical foundation appearing across the corpus, particularly in studies concerned with the risks of over-automation. Lan and Zhou's (2025) qualitative systematic review documents AI's capacity to support the forethought, performance, and reflection phases of Zimmerman's SRL model, but also highlights the distinction between human-centred and AI-centred agency configurations, noting that different configurations produce different SRL outcomes. Sargsyan (2025) draws on SRL theory to argue that agentic assessment systems must be designed to foster self-evaluation and metacognitive engagement, not merely to deliver scores. Msafiri et al. (2025) extend this by examining how AI-mediated environments can either strengthen or undermine collaborative self-regulation.

Bloom's taxonomy provides a third theoretical reference point, particularly for the design of agentic instructional systems. The IISc Instructor Agent framework (Simmhan & Kulkarni, 2025) explicitly aligns its weekly knowledge bases and learning outcomes with Bloom's cognitive levels, using taxonomy-grounded learning objectives to constrain and guide agent behaviour. This approach ensures that agentic instruction addresses the full spectrum of cognitive demand from recall to synthesis rather than defaulting to the lower-order interactions that language models handle most fluently.

Critical pedagogy frameworks, including Freire's emphasis on conscientisation and the importance of learner agency in the educational encounter, provide a normative counterpoint to purely technical optimisation frameworks. Roe and Perkins (2024) and Sargsyan (2025) both draw on critical perspectives to argue that agentic AI integration must preserve and enhance human agency rather than diminishing it, and that educational AI systems must be evaluated not only on efficiency metrics but on their contribution to emancipatory educational goals. This critical dimension is under-represented in the empirical literature, which tends to privilege quantitative performance outcomes, and represents a significant gap in current theoretical engagement with the field.

4.2. Benefits and opportunities

The benefits of agentic AI integration in higher education, as documented across the 48 included studies, cluster around three interconnected dimensions: scalability, personalisation, and operational efficiency. Fig. 4 summarises these benefits alongside the principal challenges discussed in Section 4.3.

Scalability is perhaps the most structurally significant benefit. Higher education systems globally face the challenge of expanding access while containing costs, a tension that traditional instructional models struggle to resolve. Agentic tutoring systems, administrative agents, and research support tools all operate at scale without proportional increases in human resource requirements. The ability to deploy a virtual teaching assistant that responds to student queries at any hour, or an early warning system that continuously monitors thousands of student data streams, fundamentally changes the staffing economics of student support. Agent Laboratory's research workflow automation demonstrates that tasks previously requiring significant researcher time can be substantially accelerated, with direct implications for the productivity of academic research enterprises (Unite.AI, 2025).

Personalisation at scale, long an aspirational goal for educational technologists, becomes operationally achievable through agentic AI. As documented in Section 3.2, adaptive tutoring systems have demonstrated performance improvements of 15–25% and engagement increases of up to 40% (Yuensook et al., 2025). The shift from static curriculum delivery to dynamic, learner-responsive instruction represents a qualitative change in the student experience, particularly for learners who have historically been poorly served by one-size-fits-all approaches: first-generation students, students with learning differences, and those studying in a second language.

The shift from reactive to proactive support constitutes one of the most profound opportunities identified in the literature. Traditional educational support systems are structurally reactive: advisors respond when students make appointments, early intervention occurs after attendance or grade data trigger concern, and feedback is provided after submission deadlines. Agentic systems invert this logic, continuously monitoring indicators and intervening before problems become entrenched. Proactive AI support agents, like those deployed by Element451 and Ivy Tech, demonstrate that early identification and targeted outreach can substantially improve retention and achievement outcomes.

Operational efficiency gains across assessment, administration, curriculum, and research are documented throughout the corpus. The redirection of instructor time from procedural grading to high-value mentoring and conceptual engagement, the automation of administrative workflows that currently consume significant professional capacity, and the acceleration of research processes all represent genuine contributions to institutional productivity.

4.3. Risks, challenges, and critical concerns

The risks and challenges of agentic AI integration in higher education are as significant as its potential benefits, and the reviewed literature addresses them with considerable analytical depth. Four categories of concern emerge consistently.

The risk of cognitive dependency and the erosion of deep learning is the most extensively discussed challenge. Sargsyan (2025) argues that when students interact passively with automated tutoring and feedback systems, without the metacognitive engagement that challenging, reflective learning requires, they may receive the surface appearance of support while the deeper cognitive work is performed by the agent rather than the learner. Empirical support for this concern appears in Sargsyan's (2025) finding that passive learners in agentic tutoring environments underperformed students in active learning conditions by 8.7%, a finding that underscores the importance of learning design choices over raw technological provision. This risk is not intrinsic to

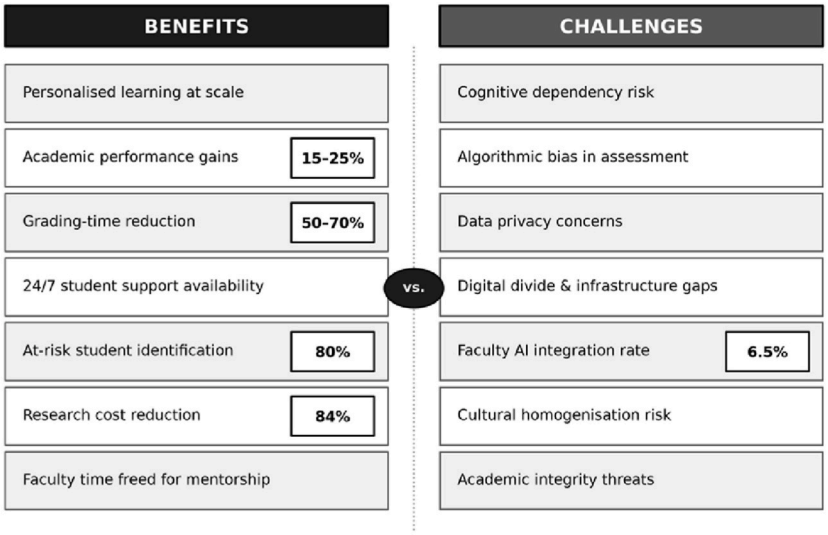


Fig. 4. Benefits and challenges of agentic AI in higher education. A side-by-side comparison of the principal benefits reported across the included studies (personalised learning at scale, academic performance gains, reduced grading time, continuous student support, at-risk identification, and research cost reduction) and the principal challenges (cognitive dependency, algorithmic bias, data privacy, the digital divide, low faculty integration, cultural homogenisation, and academic integrity threats). Indicative percentages are those reported by individual studies within the sample; benefits and challenges are not mutually exclusive.

agentic AI; it arises when these systems are deployed without deliberate design for active student engagement and metacognitive development. Lan and Zhou (2025) similarly note that AI-centred agency configurations, in which the AI makes most learning decisions autonomously, produce weaker SRL outcomes than human-centred configurations in which students retain meaningful decisional authority.

Cultural homogenisation and epistemic hegemony represent a concern that is particularly acute for globally diverse higher education systems. Agentic AI systems are predominantly trained on English-language, Western-produced data, and their embedded assumptions about knowledge, argumentation, academic register, and cultural convention tend to reflect those origins. Sargsyan (2025) documents how AI tools in language education can marginalise non-Western rhetorical traditions and penalise linguistic features associated with non-English academic cultures. Khoza and van der Walt (2025) examine this dynamic through a Global South lens, documenting how AI-enhanced pedagogies can reproduce and amplify existing inequities when their design does not account for the epistemological diversity of students, the infrastructural constraints of under-resourced institutions, or the cultural contexts in which learning takes place. Cultural homogenisation in content and assessment risks accelerating the linguistic marginalisation and epistemic hegemony already observed in international higher education.

The digital divide and infrastructure gaps constitute a structural barrier that shapes who benefits from agentic AI and who does not. Khoza and van der Walt (2025) document markedly lower institutional readiness in under-resourced regions of the Global South relative to well-resourced North American and European institutions. Where reliable internet connectivity, institutional data systems, and basic hardware are absent, agentic AI systems cannot be deployed at all, and the risk is that their proliferation in well-resourced contexts widens existing educational inequalities. The finding that only 6.5% of faculty in Armenia reported direct use of AI for educational applications (Atoyan et al., 2025, as cited in Sargsyan, 2025) illustrates the gap between technology availability and genuine pedagogical integration. The wider literature synthesised by Sargsyan (2025) likewise reports that a large proportion of educators have received insufficient preparation for AI integration, representing a workforce development challenge of considerable scale.

4.4. Governance and ethical considerations

The governance and ethical dimensions of agentic AI in higher education constitute one of the most critically important and underdeveloped areas in the reviewed literature. Six interconnected concerns emerge.

Data privacy and the governance of student data represent the most immediate and legally consequential concern. Agentic systems, particularly those performing personalised tutoring, adaptive assessment, and early warning functions, require continuous access to detailed student data: learning behaviours, performance trajectories, engagement patterns, psychological states, and in some systems, biometric data. The collection, storage, processing, and use of this data must comply with data protection legislation (including GDPR in European contexts and FERPA in the United States), but the reviewed literature suggests that institutional data governance frameworks frequently lag behind deployment practice (Kasarci et al., 2025). Sargsyan (2025) warns against unregulated collection and use of sensitive student data, noting that the granularity of data generated by agentic systems substantially exceeds that of prior educational technologies and creates correspondingly greater risks of breach, misuse, and surveillance.

Algorithmic bias, as documented in Section 3.3 and Section 4.3, operates at multiple levels: in training data, in model architecture, in evaluation criteria, and in the populations on which systems are tested. Kasarci et al. (2025), in their systematic framework for managing AI ethics in higher education, identify bias as a systemic rather than incidental problem requiring institutional-level mitigation strategies including bias audits, diverse dataset requirements, transparent algorithmic documentation, and stakeholder involvement in system design. The EDUCAUSE ethical guidelines (2025) similarly emphasise that nondiscrimination and fairness require ongoing audit processes rather than one-time technical fixes.

Transparency and explainability are governance requirements that take on specific urgency in high-stakes educational contexts. When an

agentic assessment system grades student work, or an early warning system flags a student as at-risk, affected students and their educators have a legitimate interest in understanding how that determination was made. The concept of inspectable memory traces, proposed by Acharya et al. (2025) as a design requirement for trustworthy agentic tutoring systems, points towards a broader principle: that agentic AI in education must be designed with auditability as a first-class requirement, not an afterthought. The governance framework illustrated in Fig. 5 maps these requirements across five nested governance layers: institutional strategy and leadership; policy and regulatory framework; operational implementation; stakeholder engagement; and monitoring and evaluation.

Academic integrity represents a distinctive challenge posed by agentic AI's content-generation capabilities. As agents become capable of producing high-quality academic text, completing programming assignments, and generating research outputs, the boundaries of individual intellectual work are increasingly difficult to establish. Güneş and Liman Kaban (2025), in their Delphi study of ethical challenges in higher education AI research, document expert consensus around the need for clear institutional policies, redesigned assessments that evaluate process as well as product, and AI literacy education that enables students to engage with these systems critically and responsibly. Sargsyan (2025) advocates for explicit discussion of authorship and AI use expectations in course design, while Acharya et al. (2025) argue for assessment designs that cannot be effectively completed through agentic assistance alone. These difficulties are compounded by the stochastic, dynamic, and fluid autonomy of agentic systems: Mukherjee and Chang (2025) argue that as agents co-produce outputs through iterative, self-adapting workflows, human and machine contributions become practically inseparable, straining the attribution, authorship, and liability assumptions on which academic integrity and accountability frameworks rest. For universities, this implies that integrity policies must address not only whether AI was used, but how responsibility is allocated when agentic systems act with significant independence.

The question of institutional policy frameworks is addressed directly by Kasarci et al. (2025), who synthesise 55 peer-reviewed studies on AI ethics in higher education and develop a systematic framework for

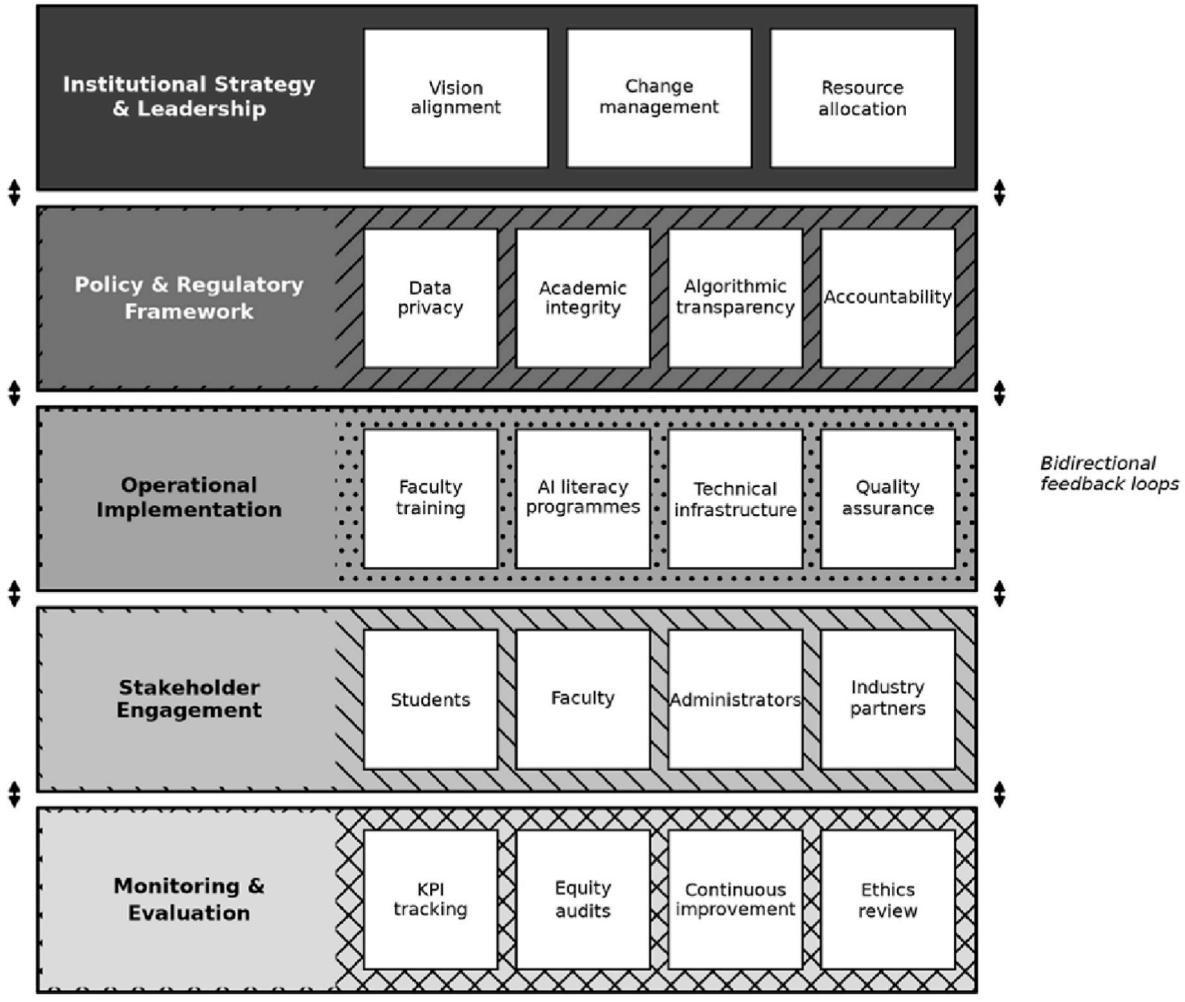


Fig. 5. Governance and ethical framework for agentic AI in higher education. A five-layer framework mapping governance responsibilities across nested institutional levels: (1) institutional strategy and leadership; (2) policy and regulatory framework, including data privacy, academic integrity, algorithmic transparency, and accountability; (3) operational implementation, including faculty training, AI literacy, and quality assurance; (4) stakeholder engagement; and (5) monitoring and evaluation. Arrows denote continuous, bidirectional feedback between layers.

institutional governance. Their framework identifies four levels of governance action: individual (student and faculty AI literacy), course (pedagogical policy and assessment design), institutional (data governance, ethics review, and deployment approval), and systemic (sector-wide regulatory standards). Hosseini and Seilani (2025) add a strategic dimension, proposing enterprise frameworks for AI adoption that begin with needs assessment and risk analysis and extend through professional development, workforce training, and continuous monitoring.

The need for adequate faculty preparation is a recurrent governance theme. The finding that a large proportion of educators globally have received insufficient AI preparation (Sargsyan, 2025) reflects a governance failure that extends well beyond a simple training gap: institutions that deploy agentic AI without ensuring that faculty understand the systems they work alongside are creating conditions for inadvertent misuse, pedagogical disempowerment, and ethical oversight failures.

4.5. Limitations of this review

Several limitations constrain the conclusions that can be drawn from this review. First, the methodological heterogeneity of the included studies, spanning experimental designs, case studies, theoretical papers, and secondary reviews, limits the strength of comparative claims and prevents formal meta-analytic synthesis. The quantitative findings reported in Section 3 represent the best available evidence but are drawn from studies with varying designs, contexts, and outcome measures. Second, publication bias may have inflated positive findings: studies reporting beneficial outcomes of agentic AI are more likely to be published than those documenting null results or implementation failures, and the field's novelty means that critical long-term evidence is structurally absent. Third, the geographic distribution of the corpus, weighted heavily towards North American and East Asian contexts, limits the generalisability of findings to Global South institutions, where infrastructure constraints, faculty preparation levels, and student demographics differ substantially. Fourth, the rapid pace of development in agentic AI means that some specific systems discussed in included studies may have evolved significantly since publication, and the boundary between current research and obsolete description is difficult to maintain. Fifth, as noted in Section 2.4, the reliance on English-language publications excludes potentially significant research conducted in other languages, further limiting geographic representativeness. Sixth, the field is expanding so rapidly that any fixed-window corpus dates quickly: recent bibliometric mapping confirms a steep post-2023 acceleration in artificial-intelligence research in higher education (Mena Guacas et al., 2026), and newer regional studies published during revision (for example, Potluri et al., 2026, on Kazakhstani students) indicate that the evidence base is broadening faster than a fixed-window review can capture. Finally, the review was not prospectively registered (for example, in PROSPERO), which we acknowledge as a limitation.

5. Conclusions and future directions

5.1. Summary of principal findings

This systematic review has synthesised evidence from 48 peer-reviewed studies published between 2023 and early 2026 to address four research questions concerning the roles of agentic AI in higher education, its theoretical foundations, reported benefits and risks, and governance considerations.

In response to RQ1, six discrete roles were identified: personalised learning and adaptive tutoring (the most extensively evidenced role, with documented academic performance gains of 15–25% and engagement improvements of up to 40%); automated assessment and feedback (with 90–95% human grader agreement and 50–70% grading time reductions, alongside significant concerns about bias and metacognitive erosion); teaching assistance and augmentation (spanning virtual TA deployment to full agentic course instruction, as demonstrated at IISc); administrative and student support (including early warning systems with predictive models reaching approximately 80% accuracy in the Ivy Tech deployment (Google for Education, 2025) and multi-agent enrolment management); curriculum design and workforce alignment (with AI-driven programmes demonstrating 50–70% operational efficiencies in financial sector analogues); and research support and academic operations (with Agent Laboratory demonstrating an 84% reduction in research costs).

In response to RQ2, the principal theoretical foundations underpinning agentic AI integration are Vygotsky's sociocultural theory and the ZPD, self-regulated learning theory, cognitive load theory, Bloom's taxonomy, and critical pedagogy frameworks. The most sophisticated implementations, such as the LEA framework, integrate multiple theoretical traditions into unified agentic architectures.

In response to RQ3, the literature documents substantial benefits in scalability, personalisation, and operational efficiency, alongside significant risks including cognitive dependency, cultural homogenisation, digital divide amplification, and inadequate faculty preparation. The finding that passive engagement with agentic systems produced underperformance of 8.7% compared to active engagement conditions (Sargsyan, 2025) is among the most policy-relevant quantitative findings in the corpus.

In response to RQ4, governance and ethical concerns centre on data privacy, algorithmic bias, transparency and explainability, academic integrity, and the need for systemic institutional policy frameworks. The evidence consistently indicates that governance development has lagged substantially behind deployment practice.

5.2. A tripartite framework for responsible integration

The synthesised evidence supports a tripartite framework for the responsible integration of agentic AI in higher education, paralleling the recommendation advanced by Sargsyan (2025) and elaborated across the governance literature. The three pillars are AI literacy, ethical infrastructure, and competence-based educator training.

AI literacy, directed at both students and staff, is a prerequisite for meaningful engagement with agentic systems rather than passive consumption of their outputs. AI literacy must encompass not only technical understanding of how agentic systems work but critical capacity to interrogate their assumptions, limitations, and implications. Students who understand what an adaptive tutoring agent is doing, and why it is doing it, are better positioned to engage with it as active learners rather than passive recipients. Faculty who understand the capabilities and constraints of agentic assessment tools are better positioned to design assessments that both exploit and safeguard against those capabilities.

Ethical infrastructure refers to the institutional governance mechanisms, data protection policies, algorithmic accountability frameworks, and academic integrity policies that must be in place before agentic AI is deployed at scale. The evidence reviewed in Section 4.4 consistently indicates that governance development has lagged substantially behind deployment practice, creating institutional risk and potential harm to students. Kasarci et al.'s (2025) four-level governance framework provides a practical template for institutions seeking to systematise their ethical oversight.

Competence-based educator training is the third pillar. Given that a large proportion of educators globally have received insufficient AI preparation, and that only 6.5% of faculty in some national contexts report direct use of AI for educational applications (Atoyan et al., 2025, as cited in Sargsyan, 2025), the professional development challenge is substantial. Training must move beyond tool familiarisation to equip educators with pedagogical frameworks for designing agentic AI-integrated learning experiences that preserve and enhance human agency.

5.3. Future research directions

The evidence base identified in this review is characterised by methodological youth: there is a preponderance of short-term case studies and design demonstrations, a scarcity of randomised controlled trials, and a near-complete absence of longitudinal studies tracking learning outcomes and institutional impacts over extended periods. The field urgently needs longitudinal empirical research that follows cohorts of students through multi-year programmes, tracks the long-term effects of agentic tutoring on academic achievement and self-regulated learning capability, and documents institutional implementation trajectories over time.

Multi-agent orchestration systems, in which specialist agents for tutoring, assessment, administrative support, curriculum oversight, and research assistance are coordinated within unified campus-wide architectures, represent the emerging frontier of deployment. Currently, the evidence base treats individual roles largely in isolation. Research that examines the systemic effects of integrated multi-agent campus deployments, their interactions, emergent properties, and cumulative impacts on student experience and institutional culture, is essential for understanding the higher-order implications of agentic AI in higher education.

To make these trajectories concrete, consider a plausible scenario five years from now. A university might operate an integrated, human-supervised multi-agent ecosystem in which (1) a teaching agent designs personalised learning pathways for each student; (2) an assessment agent grades work and returns metacognitively framed feedback; (3) a student-support agent identifies at-risk learners and coordinates timely intervention; and (4) a research agent assists staff and postgraduate students with literature synthesis, protocol drafting, and data analysis. These agents would exchange information through shared, privacy-preserving data layers under continuous human oversight, with faculty retaining authority over pedagogical and ethical decisions. Coordinating such agents safely, aligning them with institutional values, and preventing the erosion of human judgement will be among the central challenges of the coming years (Liu et al., 2026; Sudarshan et al., 2026).

Equity-focused research, examining the differential impacts of agentic AI on students from diverse linguistic, cultural, socioeconomic, and geographic backgrounds, is critically needed. The evidence reviewed in Sections 3.5 and 4.3 suggests substantial risks of inequity amplification, but the empirical base for understanding these dynamics in specific HE contexts remains thin. Research embedded in Global South institutions, where infrastructure constraints and epistemological diversity are most salient, is particularly important.

Finally, the governance and ethics dimension requires empirical investigation, not merely normative theorising. Studies that evaluate the effectiveness of specific institutional governance frameworks, compare the equity outcomes of different data governance regimes, or assess the impact of AI literacy interventions on student and faculty engagement with agentic systems would substantially strengthen the evidence base for responsible AI integration in higher education. Recent syntheses reinforce these priorities: Liu et al. (2026) show that generative-AI agent research remains concentrated in a limited set of roles and regions, and Irwansyah (2026) reframes artificial-intelligence literacy as a communication-centred practice, underscoring the need for literacy interventions that travel across linguistic and cultural contexts.

The emergence of agentic AI marks a pivotal moment for higher education, representing the transition from task-level automation to the creation of intelligent, collaborative ecosystems capable of managing significant dimensions of the educational enterprise. The ultimate measure of this transition is educational outcome rather than technological sophistication: whether agentic systems, appropriately governed and pedagogically grounded, serve to widen educational opportunity, deepen learning, and strengthen the capabilities of human educators to do what only human educators can do.

CRediT authorship contribution statement

Simon Baradziej: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing.

Data availability statement

This study is a systematic review of previously published literature. All data analysed are contained within the articles cited in the References. The data-extraction and coding materials generated during the review are available from the corresponding author on reasonable request.

Declaration of the use of generative AI and AI-assisted technologies

During the preparation of this work, the author used ChatUiT (a university-provided generative-AI assistant) in order to proofread and improve the styling and language of the manuscript. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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Declaration of competing interests

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